MIL-F-48168(PA)
12 November 1973
SUPERSESSION DATA
See Section 6

MILITARY SPECIFICATION

FLECHETTE, THIRTEEN GRAIN

1. SCOPE

1.1 This specification covers one type of flechette designated as Flechette, Thirteen Grain.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	-Ammunition and Special Weapons, General Specification for

STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-Quality Assurance Terms and Definitions
MIL-STD-1168	-Lot Numbering of Ammunition
MIL-STD-1169	-Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	-Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1315

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DRAWINGS

ARMY

9219980 -Flechette

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Flechette.-The flechette shall comply with all requirements specified on drawing (dwg.) 9219980 and with all requirements specified in applicable specifications.

3.3 Workmanship.-All flechettes shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, flash, splits and cracks. They shall be clean and free of scale, dirt, grease and other foreign matter prior to protective coating. The cleaning method used shall not be injurious to any parts nor shall the parts be contaminated by the cleaning agent.

3.4 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and

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services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of Product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawings and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission.-The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 25 completely finished flechettes which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment utilizing a typical die set which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production.

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Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any assembly or component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component to comply with any of the stated requirements.

4.3 Quality Conformance Inspection

4.3.1 Lot Formation.-The term "inspection lot", as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods and the same die set, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in Standard MIL-STD-105. Changes to the process, specifications or drawings, not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type

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of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of Major and Minor Defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each Major Defect and an AQL of 0.65 percent for each Minor Defect. When this method is elected, it shall be applied to all of the Major and Minor Defects listed in the applicable subparagraph of this section and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Flechette, Thirteen Grain (see dwg. 9219980).

Categories	Defects	Method Of Inspection	Code No.
Major:	AQL 4.0 percent		
101.	Area of a pair of opposite fins, location of centroid and width across fins at widest point.....	Gage	01001
102.	Point missing or deformed.....	Visual	01002
Minor:	AQL 10.0 percent		
201.	Total length.....	Gage	01003
202.	Straightness of shank.....	Gage	01004
203.	Positional location of fins.....	Gage	01005
204.	Evidence of poor workmanship (see 3.3).....	Visual	01006

4.3.3 Hardness (see dwg. 9219980)-Major Defect-Code No. 01007.-The contractor shall verify compliance with the applicable requirement by introducing into the heat treatment process, specimens prepared in accordance with 4.4.1. The method of inspection shall be a commercial hardness tester.

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4.3.3.1 Batch type process.-A minimum of four specimens shall be introduced into each oven batch to represent the extreme conditions which may exist within the oven. After final heat treatment, the specimens shall be tested for hardness. If any sample fails to comply with the applicable requirements, the lot shall be rejected.

4.3.3.2 Continuous type process.-A minimum of two specimens shall be introduced into the oven at intervals of not more than one hour. After final heat treatment, the specimens shall be tested for hardness. If any specimen fails to comply with the applicable requirement, the quantity of flechettes heat treated since the last acceptable sample shall be rejected.

4.3.4 The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.3.2 and 4.3.3, see 6.3.

4.4 Test Methods and Procedures

4.4.1 Hardness.-Specimens shall be prepared from the same heat of wire as being used in the manufacturing process by grinding flats sufficiently wide to allow a valid hardness reading to be taken.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level C.-Packing and marking shall be in accordance with Specification MIL-STD-1169.

6. NOTES

6.1 Ordering Data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data Cards - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for the submission of first article sample.

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6.2 Inspection Code Numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types-Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial

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equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423).

The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Supersession data.-This specification includes the requirements of Purchase Description PA-PD-2899 dated 6 October 1966.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1315-A-020

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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